

ASIC Tools Makefile Guide

Creating a project:

1. Ensure scripts/ is populated and in same folder as Makefile
2. run “make proj PROJECT=<project_name>”
3. cd into project_name
4. run “make logs”
5. begin work
6. run “make help” if you get lost

```
ubuntu@asic:~/workspace/cmp_32$ make help

Linting
lint to lint all your rtl

Verilator Simulation (0,1)
sim to simulate all testbenches with Verilator
sim/ to simulate one testbench with Verilator

Icarus Simulation (1, 0, X, Z)
isim to simulate all testbenches with Icarus Verilog
isim/ to simulate one testbench with Icarus Verilog

Icarus Gate-Level Simulation (1, 0, X, Z)
glsim/ to simulate your design with gate-level simulation with Icarus

Synthesis
synth runs synthesis with Yosys 0.51
requires synth.js to be configured in scripts/

OpenSTA
sta
requires sta.tcl to be configured in scripts/
run "make openlane_sta" before trying this

Openlane
openlane runs the OpenLane 2.0 Classic Flow
requires config.yaml or config.json in the project home

RARS
rars opens RARS
requires rars.jar in asm/

Log all relevant info about tools versions and pdk paths
log loads .logs with relevant info

Remove build artifacts and logs
clean removes tool generated output

Project Generation
proj
pass PROJECT= to name your project
ex: make proj PROJECT=mul
```

Command List:

lint runs Verilator using --lint-only flag

lint_top TOP_MODULE=design TOP_FILE=design.sv

sim runs Verilator against all the testbenches in sim/

isim runs Icarus against all the testbenches in sim/

sim/<testbench folder > runs Verilator against the specific testbench

isim/<testbench folder > runs Icarus against the specific testbench

glsim/<testbench folder > runs Icarus gate-level sim (post-flow only)

cocotests runs your tests in cocotests/ (for cocotb only)

synth runs the “synth.js” yosys script (implement the script in scripts/)

sta runs the “sta.tcl” OpenSTA script (implement the script in scripts/)

openlane runs the entire OpenLane 2.0 Classic flow (needs config file)

openlane_sta runs the flow through step 13 - floorplan (needs config file)

openroad opens the OpenROAD GDSII in your most recent OpenLane run

list-versions lists the tool-versions to tool-version.log

list-libs lists the Liberty timing files of the standard cell library to pdk_libs.log

list-spice list the spice models of the pdk to pdk_spice.log

list-magicrc list the magicrc to pdk_magicrc

logs runs all of the list- commands at once

clean removes build artifacts from simulation, synthesis, and logs

help prints the list of commands with descriptive comments on usage

proj PROJECT=<project_name> will create a directory called project_name/ that will contain the Makefile, a populated rtl, scripts, and sim folder, and an empty logs, synth, and asm subfolders. Inside rtl/ will be an empty project_name.sv file. Inside tests/ will be a folder called tb_project_name/ which contains an empty file named tb_project_name.sv. Not defining the PROJECT variable will result in the default <project_name> being called “project”